Bambu Lab A1 3D Printer

WHAT'S IN THE KIT

- Bambu A1 Combo (with AMS lite multi-color unit)
- 4 colors of filament
- SD card for file transfer
- Toolkit

QUICK-START GUIDE

1. Level-free setup: the A1 auto-calibrates on startup.
2. Students design in Tinkercad (free, browser) and export STL files.
3. Slice in Bambu Studio, save to SD card, and print.
4. First print: something small (~20 min) so students see the full cycle.
5. Keep the printer where students can safely watch — it's the best advertisement for design iteration.

FULL LESSONS IN THIS GUIDE

PRO TIP: *Multi-color printing via the AMS is a showstopper — but single-color prints are 2-3x faster for classwork.*

Bambu Lab A1 3D Printer

LESSON · GRADES 3-5

Design a Keychain

LEARNING OUTCOMES

- Students will design a 3D model with basic CAD operations
- Students will prepare a design within print constraints
- Students will describe how a printer builds objects layer by layer

MATERIALS

- Bambu A1 printer (visible to the class)
- Chromebooks with tinkercad.com
- Keychain rings for finished prints
- Size constraint card (e.g., 50mm max, 5mm thick)

INSTRUCTIONS

1. Show a print starting — the first layers going down hook every kid in the room.
2. Tinkercad basics: drag shapes, resize, group; make letters stand on a base.
3. Students design personal keychains within the size constraint.
4. Teacher batch-checks: floating parts and paper-thin features get flagged for revision.
5. Print in batches across days; students watch their own print when possible.
6. Attach rings; reflect: what would you change in version 2?

ACTIVITY

> Design a Keychain — every student designs and receives a real printed object. 'I made this' with a thing in their pocket outperforms any lesson about technology.

1 · A1 Combo

Bambu Lab A1 3D Printer

LESSON · GRADES 6-8

Problem-Solvers Print Shop

LEARNING OUTCOMES

- Students will identify a real problem suited to a printed solution
- Students will design with measurements taken from the real world
- Students will install and evaluate a solution in place

MATERIALS

- Bambu A1 printer + filament
- Chromebooks with Tinkercad
- Calipers or rulers for real measurements
- Problem-hunting walk sheet

INSTRUCTIONS

1. Problem hunt: walk the classroom logging small physical annoyances (wobbly table, tangled cords, unlabeled bins).
2. Teams claim one problem and measure the real constraint precisely.
3. Design in Tinkercad with those measurements — fit is the whole game.
4. Print v1; test-fit immediately. Most first fits fail: measure again, revise, reprint small.
5. Install working solutions where they belong.
6. Two weeks later: which fixes survived actual use? Why?

ACTIVITY

> Problem-Solvers Print Shop — find a real problem, measure it, print the fix, install it. The two-week survival check separates engineering from decorating.

1 · A1 Combo

Bambu Lab A1 3D Printer

LESSON · GRADES 9-12

Client-Based Design Brief

LEARNING OUTCOMES

- Students will gather requirements from a real client
- Students will manage a design-review-revise cycle professionally
- Students will deliver and document a fabricated product

MATERIALS

- Bambu A1 printer + filament
- CAD access (Tinkercad or Onshape)
- Client interview template
- Design review sign-off sheet

INSTRUCTIONS

1. Recruit clients: teachers, the front office, a younger class — anyone with a small physical need.
2. Teams run client interviews: need, constraints, must-haves vs. nice-to-haves.
3. Concept review: client approves a sketch before CAD begins.
4. CAD review: client sees the model render and signs off (or requests changes).
5. Print, deliver, and collect written client feedback.
6. Portfolio entry: the full project record from interview to delivery.

ACTIVITY

Client-Based Design Brief — real clients, sign-off gates, and delivered products. Managing a client who changes their mind is the most authentic engineering lesson available.